

Day 4: Authentication Verification (SPF/DKIM/DMARC) & Domain Intelligence

Key Milestone: Email Authentication Verification Engine

Sprint Focus: Execution of core deliverables for Day 4 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed: SPF/DKIM Verification Subsystem, Typosquatting Detection Algorithm, Interactive Nodal Canvas.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership: Explainable AI, Intent & Phishing Detection, Social Engineering signals | **Key Modules:** NLP/AI Engine, Explainability, Content Risk scoring

Day 4 Execution: Added sentiment analysis and social engineering urgency cue detection (fear, scarcity, financial inducement, time pressure).
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership: Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis | **Key Modules:** Header Forensics, Mail Path, Infrastructure Risk

Day 4 Execution: Engineered cryptographic signature validator for DKIM (``rsa-sha256``) and TXT record parser for SPF mechanisms (``ip4``, ``include``, ``all``).
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership: URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection | **Key Modules:** IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 4 Execution: Implemented lookalike domain detection (typosquatting, homoglyph replacement, sub-domain tunneling) against Alexa Top 10K brands.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership: Attack Relationship Graph, Similarity detection, Campaign grouping & correlation | **Key Modules:** Graph Engine, Campaign Detection, Similar Attack Detection

Day 4 Execution: Built graph edge weighting logic: connecting disparate emails sharing identical originating IP subnets (/24 CIDR) or registrant names.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership: Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions | **Key Modules:** Risk Scoring, Case Management, Response Recommendations

Day 4 Execution: Implemented threat level classification logic (``VERIFIED CLEAN``, ``LOW``, ``MEDIUM``, ``HIGH``, ``CRITICAL THREAT``) with actionable remediation steps.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership: Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX | **Key Modules:** Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 4 Execution: Implemented the interactive Attack Infrastructure Graph visualization using HTML5 Canvas and SVG nodal physics.

System Integration & Quality Verification

Cross-Module Integration: Verified SPF/DKIM validation against real-world test emails with passing, soft-fail, and forged signatures.

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 4 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 4 MILESTONE SIGNED OFF